File

Name: R5-fit-A2-T500
Description: KISSsoft example
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1 Overview

Calculation method: According to DIN 7190-1:2017, elastic range
with additions such as centrifugal force, micro sliding, mounting, etc.
for a horizontal or longitudinal interference fit

Diameter shaft (mm)	[DiA]/[DiI]	50.00	/ 0.00		
Diameter hub (mm)	[DaA]/[DaI]	100.00	/50.00		
Length of Interference fit (mm)	[l]	40.00			
Diameter of joint (mm)	[DF]	50.00			
with special allowances:					
Upper allowance Shaft (µm)	[AoI]	75.0			
Lower allowance Shaft (µm)	[AuI]	75.0			
Tolerance measure Shaft (µm)	[TI]	0.0			
Upper allowance Hub (µm)	[AoA]	0.0			
Lower allowance Hub (µm)	[AuA]	0.0			
Tolerance measure Hub (µm)	[TA]	0.0			
Largest interference (µm)	[Po]	75.0			
Smallest interference (µm)	[Pu]	75.0			
Nominal torque (Nm)	[T]	500.00			
Application factor	[KA]	1.00			
Service torque (Nm)	[Tb]	500.00			
Axial force (N)	[FA]	0.00			
Bending moment (Nm)	[Mb]	0.00			
Radial force (N)	[Fr]	0.00			
Circumferential force (N)	[Fu]	20000.00			
Speed (1/min)	[n]	0.00			
Interference (µm)	[P]	75.0	(75.0	..	75.0)
Embedding (µm)	[s]	4.80			
Effective interference at 20°C(68°F) (µm)	[Pw]	70.20	(70.2	..	70.2)
Effective interference at working temp.. (µm)	[PwTh]	70.20	(70.2	..	70.2)
Service temperature shaft (°C)	[ThB]	20.0			
Service temperature hub (°C)	[ThB]	20.0			
Pressure stress by					
- Interference (after mounting) (N/mm²)	[pM]	113.20	(113.20	..	113.20)
- Interference (working) (N/mm²)	[p]	113.20	(113.20	..	113.20)
- Bending moment (N/mm²)	[pb]	0.00			
- Radial force (N/mm²)	[pr]	0.00			
Coefficient of friction, longitudinal	[µa]	0.120			
Coefficient of friction, circumferential	[µu]	0.120			
Safety against sliding	[Sr]	4.27	(4.27	..	4.27)
Required safety against sliding	[SSr]	1.50			

Equivalent stress according to von Mises
$\sigma V = (\sigma R^2+\tau T^2-\sigma R \cdot \tau T)^{0.5}$

1.1 Shaft

Material, own Input		C45 (1)			
Young's modulus (N/mm²)	[El]	215000.00			
Poisson's ratio (-)	[vl]	0.30			
Density (kg/m³)	[ρ]	7830.00			
Coefficient of thermal expansion (10^-6/K)	[α]	11.50			
Tensile strength (component's raw diameter) (N/mm²)	[Rm]	650.00			
Yield point (component's raw diameter) (N/mm²)	[Rp]	200.00			
Component raw diameter (mm)	[deff]	55			
Surface class of roughness	Own Input				
Surface roughness (µm)	[Rz]	6.00			
External diameter (mm)	[DiA]	50.00			
Diameter increase during operation (µm)	[ΔD]	-18.43	(-18.43	..	-18.43)
Diameter increase during operation (o/oo)	[ΔD]	-0.37	(-0.37	..	-0.37)
Equivalent stress outer diameter (N/mm²)	[σVa]	113.20	(113.20	..	113.20)
- Radial stress (N/mm²)	[σRa]	-113.20	(-113.20	..	-113.20)
- Tangential stress (N/mm²)	[σTa]	-113.20	(-113.20	..	-113.20)
- With outside load (N/mm²)	[σVaMF]	113.20	(113.20	..	113.20)
Inner diameter (mm)	[Dil]	0.00			
Equivalent stress inner diameter (N/mm²)	[σVi]	113.20	(113.20	..	113.20)
- Radial stress (N/mm²)	[σRi]	-113.20	(-113.20	..	-113.20)
- Tangential stress (N/mm²)	[σTi]	-113.20	(-113.20	..	-113.20)
Safety against fracture	[SiRm]	5.74	(5.74	..	5.74)
Required safety against fracture	[SSi.Rm]	1.50			
Safety against yield point	[Si.Rp]	1.77	(1.77	..	1.77)
Required safety against yield point	[SSi.Rp]	1.10			

1.2 Hub

Material, own Input		C45 (1)			
Young's modulus (N/mm²)	[EA]	215000.00			
Poisson's ratio (-)	[vA]	0.30			
Density (kg/m³)	[ρ]	7830.00			
Coefficient of thermal expansion (10^-6/K)	[α]	11.50			
Tensile strength (wall thickness) (N/mm²)	[Rm]	700.00			
Yield point (wall thickness) (N/mm²)	[Rp]	300.00			
The hub material strength values are defined using the wall thickness as the raw diameter.					
Component raw diameter (wall thickness) (mm)	[deff]	26			
Surface class of roughness	Own Input				
Surface roughness (µm)	[Rz]	6.00			
External diameter (mm)	[DaA]	100.00			
Diameter increase during operation (µm)	[ΔD]	35.10	(35.10	..	35.10)
Diameter increase during operation (o/oo)	[ΔD]	0.35	(0.35	..	0.35)
Equivalent stress outer diameter (N/mm²)	[σVa]	75.47	(75.47	..	75.47)
- Radial stress (N/mm²)	[σRa]	0.00	(0.00	..	0.00)
- Tangential stress (N/mm²)	[σTa]	75.47	(75.47	..	75.47)
Inner diameter (mm)	[Dal]	50.00			
Diameter increase during operation (µm)	[ΔD]	51.77	(51.77	..	51.77)
Diameter increase during operation (o/oo)	[ΔD]	1.04	(1.04	..	1.04)
Equivalent stress inner diameter (N/mm²)	[σVi]	264.13	(264.13	..	264.13)
- Radial stress (N/mm²)	[σRi]	-113.20	(-113.20	..	-113.20)
- Tangential stress (N/mm²)	[σTi]	188.66	(188.66	..	188.66)
- Radial stress with outside load (N/mm²)	[σRai]	-113.20	(-113.20	..	-113.20)
- Tangential stress with outside load (N/mm²)	[σTai]	188.66	(188.66	..	188.66)
- With outside load (N/mm²)	[σViMF]	264.13	(264.13	..	264.13)

Safety against fracture	[Si.Rm]	2.65	(	2.65	..	2.65)
Required safety against fracture	[SSi.Rm]	1.50				
Safety against yield point	[Si.Re]	1.14	(	1.14	..	1.14)
Required safety against yield point	[SSi.Re]	1.10				

Mere elastic stress, no verification of elastic plastic interference fit according to DIN 7190-1.

2 Service / Mounting / Dismounting

Transverse-interference-fit:						
Mounting clearance (mm)	[PsTh]	0.050				
Temperature difference for mounting:						
Shaft temperature: (°C) Hub temperature: [ThA] (°C)						
Mount with shaft temperature 20 °C:						
Assembly temperature shaft (°C)	[θI]	20				
Assembly temperature hub (°C)	[θA]	237.391				
Mount with shaft temperature -150 °C:						
Assembly temperature shaft (°C)	[θI]	-150				
Assembly temperature hub (°C)	[θA]	111.739				
calculated using the coefficient of thermal expansion						
Sub-cooling of the shaft according to DIN 7190-1 (10^-6/K)	[αw]	8.50				
Longitudinal pressure fit:						
Assembly temperature shaft (°C)	[θM]	20.00				
Assembly temperature hub (°C)	[θM]	20.00				
Coefficient of friction, longitudinal	[μe=μa*	1.300]	0.16			
Press on (force) (kN)	[Fpress]	110.95	(110.95	..	110.95)	
Coefficient of friction, longitudinal	[μll=μa*1.600]	0.19				
Press out (force) (kN)	[Fpress]	136.56	(136.56	..	136.56)	
Coefficient of friction	[μ]	0.19				
Max. torque to avoid Micro sliding (Nm)	[Tlimit]	653.28	(653.28	..	653.28)	

Notice concerning the display: Number-1 (Number-2.. Number-3):
- Number-1: Value calculated with the mean allowance
- Number-2: Value of the smallest possible allowance
- Number-3: Value of the largest possible allowance

Notice: All strains and stresses are calculated for the purely elastic case.

End of report (lines: 154)
